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8.38 33. The gravitational acceleration at a planet's surface is 25.0 m/s^2. Find the acceleration at an altitude equal to half the planet's radius.

8.49 45. The asteroid that exploded over Chelyabinsk, Russia, in 2012 (see Application on page 155) was moving at 35.5 km/s relative to the Sun just before it entered Earth's atmosphere. Calculations based on orbital observations show that it was moving at 11.2 km/s at aphelion (its most distant point from the Sun). Find the distance at aphelion, expressed in astronomical units (1 AU is the average distance of Earth from the Sun; see Appendix E).

8.56 52. Two meteoroids are 250,000 km from Earth's center and moving at 2.9 km/s. One is headed straight for Earth, while the other is on a path that will come within 8500 km of Earth's center (Fig. 8.16). Find the speed of (a) the first meteoroid when it strikes Earth and (b) the second meteoroid at its closest approach. (c) Will the second meteoroid ever return to Earth's vicinity?

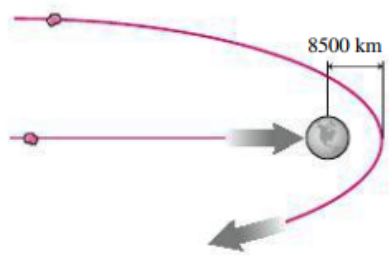

FIGURE 8.16 Problem 52

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60. Two meteoroids are 150,000 km from Earth's center and heading straight toward Earth, one at 6 km/s, the other at 12 km/s. At what speeds will they strike Earth?

64. A 750-kg spacecraft has total energy −0.57 TJ and is in circular orbit around the Sun. Find (a) its orbital radius, (b) its kinetic energy, and (c) its speed.

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